Time Series Analysis with Gretl

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Problem 1 – Forecasting Energy Variables

1.1 Data Description

The primary series is U.S. monthly net solar electricity generation (MkWh), from the EIA website, covering monthly data from January 2011 to December 2024 (168 observations). The raw series in Figure 1 shows near-exponential growth from 37 MkWh in January 2011 to 24,483 MkWh in December 2024, a 600x increase with clear seasonality. A log transformation is applied. The two other datasets included in the VAR are natural gas net generation (MkWh) and retail electricity price (cents/kWh), both log-transformed for comparison. We chose these three series because they represent the evolution of the U.S. energy sector. How they vary and how they interact is what will determine the future of energy consumption in one of the world leading countries in energy usage and production.

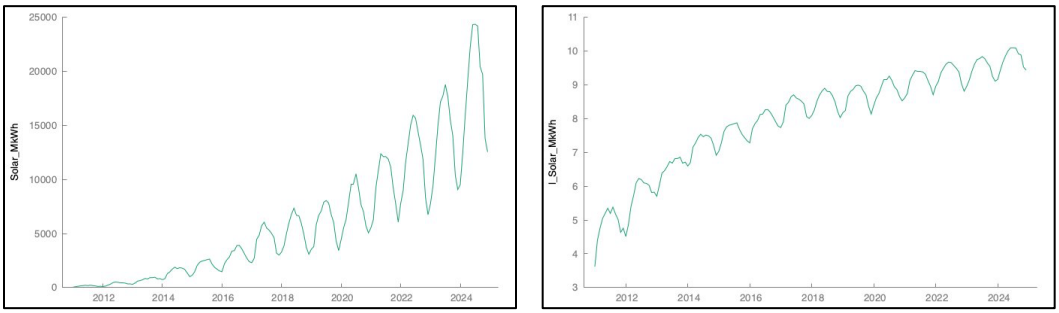


Figure 1. Solar net generation: raw series (left) and log-transformed series (right), January 2011 – December 2024.

1.2 Model Selection and Residuals

The whole dataset (n=168) is split into training (n=144) and test (n=24) sets. The model is estimated on the training sample. ARMA(p,q) combinations for $p,q \in \{0,...,4\}$ are estimated on ln(Solar) with a constant linear time trend and 11 monthly seasonal dummies, then ranked by BIC. AR(2) achieves the global BIC minimum (−211.61), outperforming the alternatives. One thing to point out is that the fitted AR(2) is technically non-stationary, since the sum of its coefficients is extremely close to 1. This can be attributed to the explosive growth trend in production seen by the solar generation over the analysed period. The model is retained given its fit and performance, with this limitation acknowledged.

The residuals chart (Figure 2) shows most ACF and PACF bars within the 95% confidence bands. Significant spikes appear at lags 8 (ACF = −0.154, p = 0.021) and 18 (ACF = 0.176, p = 0.008). This indicates some residual autocorrelation, which can still be explained by the fast acceleretion of the usage of solar energy, but overall the model is solid and is retained for the forecasting.

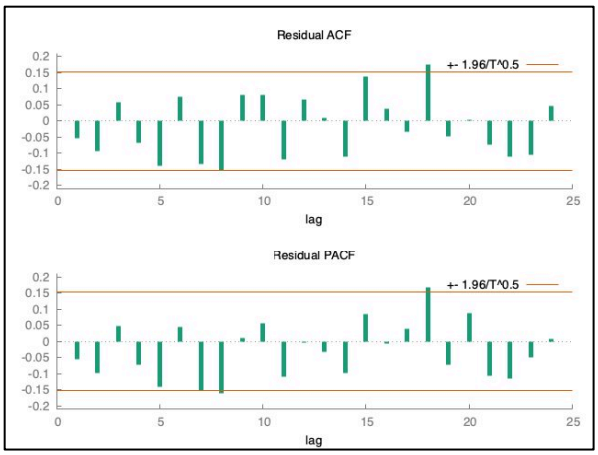


Figure 2. AR(2) residual ACF and PACF. Most bars within 95% bands.

1.3 Model Forecasting

The model is estimated on the January 2011 – December 2022 period and used to forecast the next 24 months (January 2023 – December 2024). Figure 3 shows the forecast closely tracking the actual series for the last 2 years of the observed data. RMSE = 0.044 and Theil's U2 = 0.240 confirm the strength of the model.

In the next step, the AR(2) model forecasts for 24 months out-of-sample (January 2025 – December 2026). Figure 3 shows the forecast continuing the upward trend maintaining seasonal pattern. The forecast for June 2026 is 10.49 log units, approximately 36,000 MkWh, consistent with the historical trajectory.

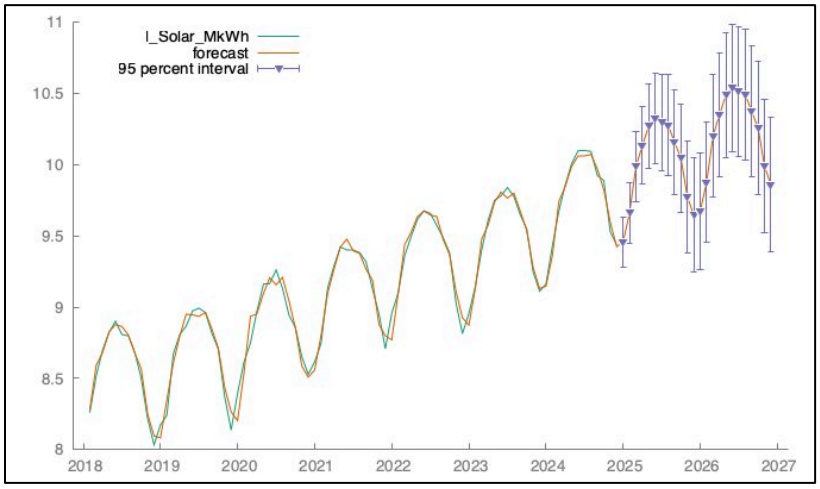


Figure 3. AR(2) out-of-sample forecast with 95% confidence interval, January 2025 – December 2026.

1.4 VAR Model

As solar expands, it should in principle replace gas production and put downward pressure on energy prices. The VAR tests for these dynamics. Looking at Figure 4, the natural gas exhibits strong seasonality and an upward trend. Electricity price was broadly stable until 2021, then rose sharply following the post-pandemic energy shock. Both series are log transformed in order to proceed.

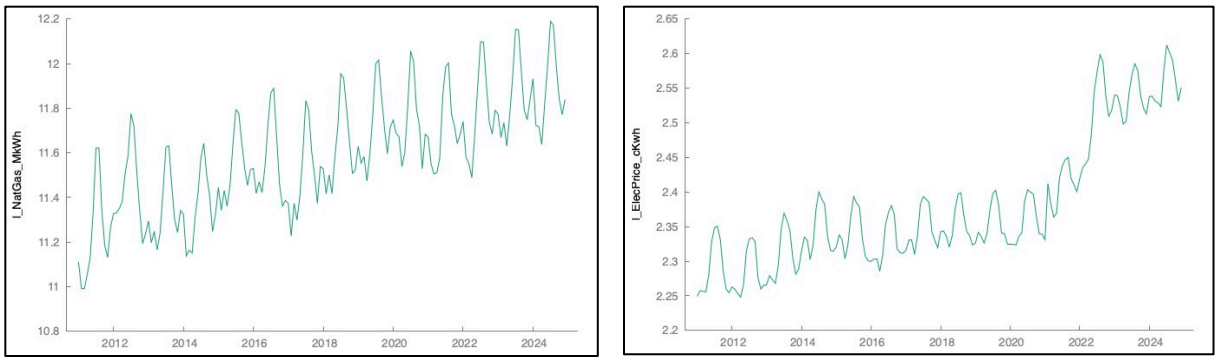


Figure 4. Log Natural gas generation (MkWh) and log retail electricity price (cent/kWh), January 2011 – December 2024.

VAR(2) is estimated. The system achieves $R^2 = 0.996$, 0.973, and 0.987 for solar, gas and price respectively. The Portmanteau test returns LB(41) = 368.16, p = 0.254, confirming adequate fit. Granger causality test has found that solar generation is not caused by either other series. Price and electricity have significant results, as price moves gas production adjusts. Ultimately gas and solar have no significant correlation. The VAR(2) residual correlogram shows once again small autocorrelations at lag 18 for solar, that we attribute to the same reasons we explained before.

1.5 Model Comparison

Figure 5 overlays the VAR(2) and AR(2) forecasts for log solar generation over January 2025 – December 2026. The two forecasts are close, with a maximum difference of approximately 0.05 log units across the 24-month horizon. Both are demonstrated to predict future performance with good accuracy. Incorporating gas production and price in VAR does not improve the performance of the model. At the end, AR(2), by a short margin, remains the preferred model.

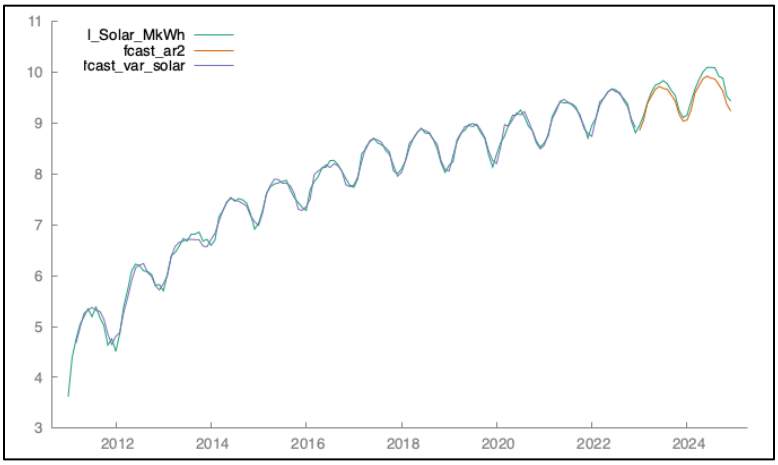


Figure 5. VAR(2) vs. AR(2) forecast for log solar generation, January 2025 – December 2026.

Problem 2 – Modelling Volatility

2.1 Data Exploration

Since we have been looking at energy, the stock we selected is NextEra Energy Inc. (NEE). NEE is the largest U.S. renewable energy utility company. The daily adjusted closing prices are sourced from Yahoo Finance over the period 11 May 2022 – 10 May 2026 (1,002 total observations). The goal of this section is to build a model to analyse volatility of this large-cap energy stock over the last 5 years of data and forecast the volatility of future instances.

Mean	0.000394177
Median	0.000728483
Minimum	-0.0940020
Maximum	0.0675426
Standard deviation	0.0172704
C.V.	43.8137
Skewness	-0.526466
Ex. kurtosis	3.70649
5% percentile	-0.0256110
95% percentile	0.0265695
Interquartile range	0.0188277
Missing obs.	1

Table 1. Summary statistics of NEE daily log-returns, 05/2022 – 05/2026.

The mean return is near zero (0.039%), which is consistent with the market at the daily frequency. Similarly, standard deviation (daily) is 1.727%. The annualised volatility of 27.41% is higher than the market, suggesting a higher sensitivity of the stock to interest rate movements, fairly consistent with the energetic sector.

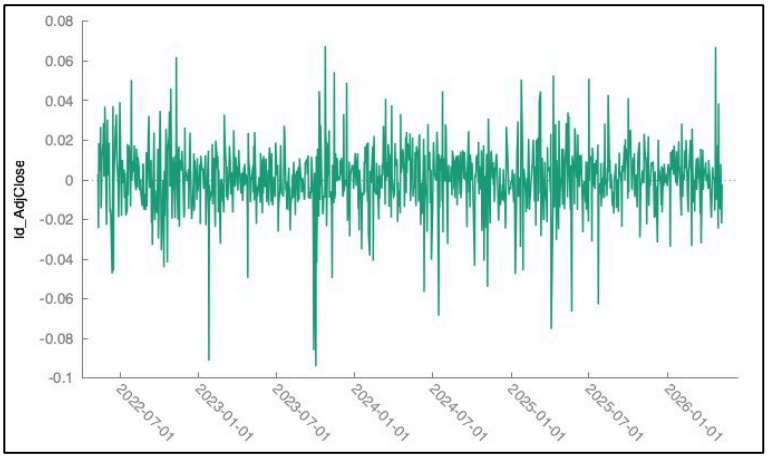


Figure 6. NEE daily log-returns, 05/2022 – 05/2026. Volatility clustering: turbulence around 2023 and calmer 2024–2025.

2.2 Testing ARCH Models

The ACF and PACF of squared excess returns (Figure 7) show significant autocorrelation at lags 2 and 3. Q-statistic rejects the null hypothesis of white noise beyond lag 3 ($p < 0.001$ at lag 30, $Q = 55.79$). This confirms that the conditional variance is serially dependent, therefore large squared returns cluster in time. The Engle ARCH-LM test (5 lags) yields $LM = 31.88$ ($p < 0.001$), rejecting the null hypothesis of no ARCH effects.

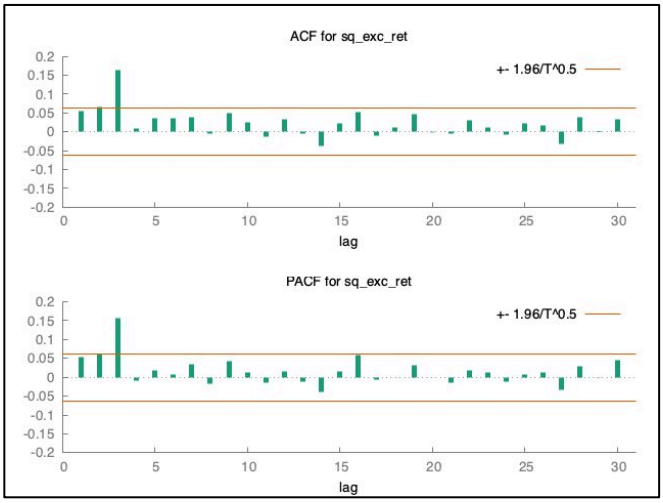


Figure 7. ACF and PACF of squared excess returns. Significant spikes at lags 2–3 confirming volatility clustering.

2.3 GARCH(1,1) Estimation

Therefore a GARCH(1,1) model is estimated. Table 2 reports the estimates. All three variance parameters are statistically significant. The likelihood ratio test yields Chi-square(2) = 37.74 ($p < 0.001$), confirming the use of a GARCH model.

```
Function evaluations: 61
Evaluations of gradient: 18

Model 2: GARCH, using observations 2022-05-11:2026-05-08 (T = 1002)
Dependent variable: exc_ret
Standard errors based on Hessian

            coefficient   std. error      z      p-value
  ---------------------------------------------------------
  const      0.000297223   0.000510830   0.5818   0.5607

  alpha(0)   9.02768e-05   3.16386e-05   2.853    0.0043  ***
  alpha(1)   0.152394      0.0660591     2.307    0.0211  **
  beta(1)    0.554476      0.147284      3.765    0.0002  ***

Mean dependent var   0.000000   S.D. dependent var   0.017270
Log-likelihood       2664.471   Akaike criterion    -5318.943
Schwarz criterion   -5294.394   Hannan-Quinn        -5309.614

Unconditional error variance = 0.000307976
Likelihood ratio test for (G)ARCH terms:
  Chi-square(2) = 37.7354 [6.39532e-09]
```

Table 2. GARCH support table.

Volatility shocks are quick, they last around 2 days. This can be explained by the structural change NEE, and almost the entire stock market, has undergone with the post-pandemic uncertainty and bounce back. High-volatility in 2022–2023 and a calmer regime registered in 2024–2025. A GARCH model with regime-switching or a longer sample would likely produce better fitting estimates. Despite this, the model passes the tests and is used for the volatility forecasting.

2.4 Residual Diagnostics

Figure 8 shows the ACF and PACF of standardised squared residuals. All autocorrelations lie within the 95% confidence lines. The Ljung-Box Q-statistic at lag 30 is 14.51 ($p = 0.992$) suggesting no lasting ARCH effects. The model fully accounts for the conditional heteroskedasticity in the residuals.

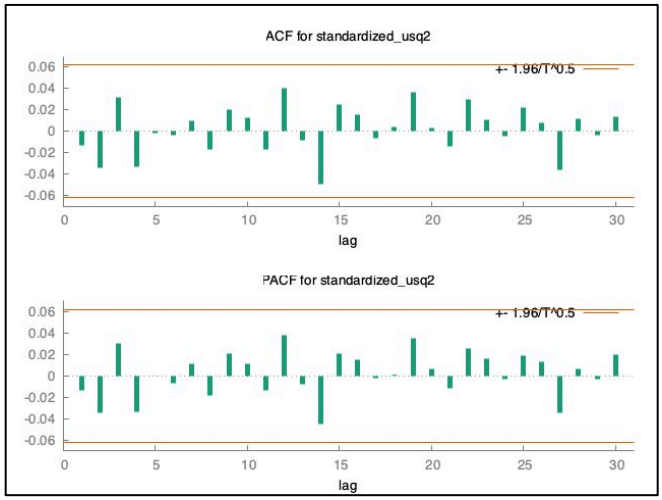


Figure 8. ACF and PACF of standardised squared residuals. All autocorrelations lie within confidence.

2.5 One-Day-Ahead Volatility Forecast

The one-day-ahead forecast for 11 May 2026 is obtained from the fitted GARCH model. The forecast return is approximately 0 (0.000297) with a predicted daily volatility forecast of 1.63% (std. error = 0.016347) which would correspond to an annualised volatility of 25.93%. The 95% forecast interval for the next-day excess return is [−3.17%, +3.23%]. 25.93% is slightly below the full-sample historical annualised volatility of 27.41%, highlighting a low persistence model. The forecast predicts a near-average volatility regime with no indication of stress for the NEE stock over the analysed period (which is a single day).

Appendix

```
? smpl 2011:01 2022:12
Full data range: 2011:01 - 2024:12 (n = 168)
Current sample: 2011:01 - 2022:12 (n = 144)

AR = 0 MA = 0 BIC = 219.83192
AR = 0 MA = 1 BIC = 113.74733
AR = 0 MA = 2 BIC = 46.496811
AR = 0 MA = 3 BIC = 7.6431525
AR = 0 MA = 4 BIC = -20.957534
AR = 1 MA = 0 BIC = -188.43255
AR = 1 MA = 1 BIC = -186.74473
AR = 1 MA = 2 BIC = -181.858
AR = 1 MA = 3 BIC = -176.88385
AR = 1 MA = 4 BIC = -181.53012
AR = 2 MA = 0 BIC = -211.6119
AR = 2 MA = 1 BIC = -203.9665
AR = 2 MA = 2 BIC = -199.1309
AR = 2 MA = 3 BIC = -194.25873
AR = 2 MA = 4 BIC = -192.4121
AR = 3 MA = 0 BIC = -205.30847
AR = 3 MA = 1 BIC = -206.31553
AR = 3 MA = 2 BIC = -205.20874
AR = 3 MA = 3 BIC = -191.61003
The convergence criterion was not met
>> arima i 0 j ; l_Solar_MkWh 0 time dm1 dm2 dm3 dm4 dm5 dm6 dm7 dm8 dm9 dm10 dm11
>> loop for j=0..4
```

Figure A1. ARMA BIC selection table.

```
Model 1: ARMAX, using observations 2011:03-2024:12 (T = 166)
Estimated using least squares (conditional ML)
Dependent variable: l_Solar_MkWh

             coefficient    std. error      z      p-value
  ------------------------------------------------------------
  const       0.0978452     0.0926737      1.056   0.2911
  phi_1       0.701692      0.0731422      9.594   8.51e-22 ***
  phi_2       0.253445      0.0702782      3.606   0.0003   ***
  time        0.000793395   0.000475898    1.667   0.0955   *
  dm1         0.187336      0.0351848      5.324   1.01e-07 ***
  dm2         0.401977      0.0397822     10.10    5.28e-24 ***
  dm3         0.577149      0.0488153     11.82    2.97e-32 ***
  dm4         0.445061      0.0535215      8.316   9.13e-17 ***
  dm5         0.394362      0.0454619      8.675   4.15e-18 ***
  dm6         0.313396      0.0452425      6.927   4.30e-12 ***
  dm7         0.212340      0.0419073      5.067   4.04e-07 ***
  dm8         0.197790      0.0387890      5.099   3.41e-07 ***
  dm9         0.0990135     0.0389057      2.545   0.0109   **
  dm10        0.0723812     0.0356478      2.030   0.0423   **
  dm11       -0.0890132     0.0353648     -2.517   0.0118   **

Mean dependent var   8.051552   S.D. dependent var   1.389902
Mean of innovations  2.40e-15   S.D. of innovations  0.087789
R-squared            0.996349   Adjusted R-squared   0.996037
Log-likelihood       176.1642   Akaike criterion    -322.3285
Schwarz criterion   -275.6486   Hannan-Quinn        -303.3808

                       Real  Imaginary    Modulus  Frequency
  ------------------------------------------------------------
  AR
    Root  1           1.0368     0.0000     1.0368     0.0000
    Root  2          -3.8055     0.0000     3.8055     0.5000
  ------------------------------------------------------------
```

Figure A2. AR(2) full sample estimation output.

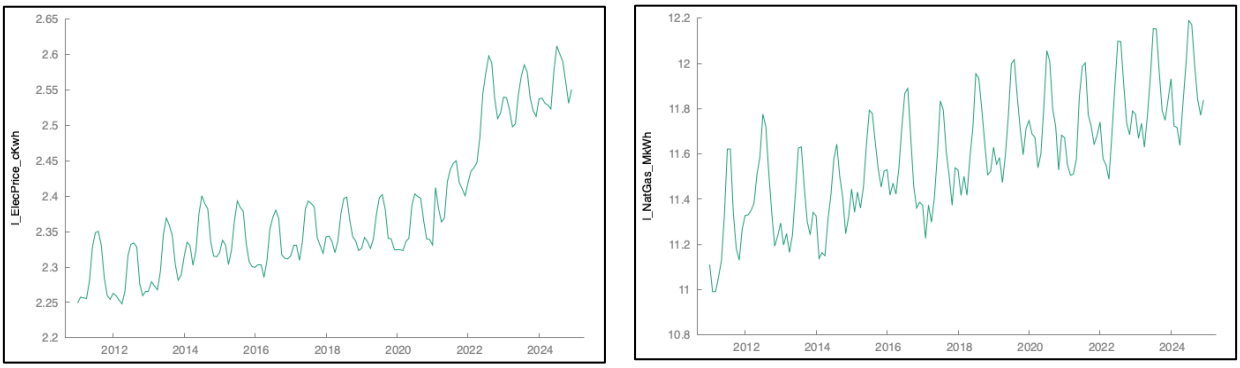


Figure A3. Log NatGas generation (left) and log electricity price (right).

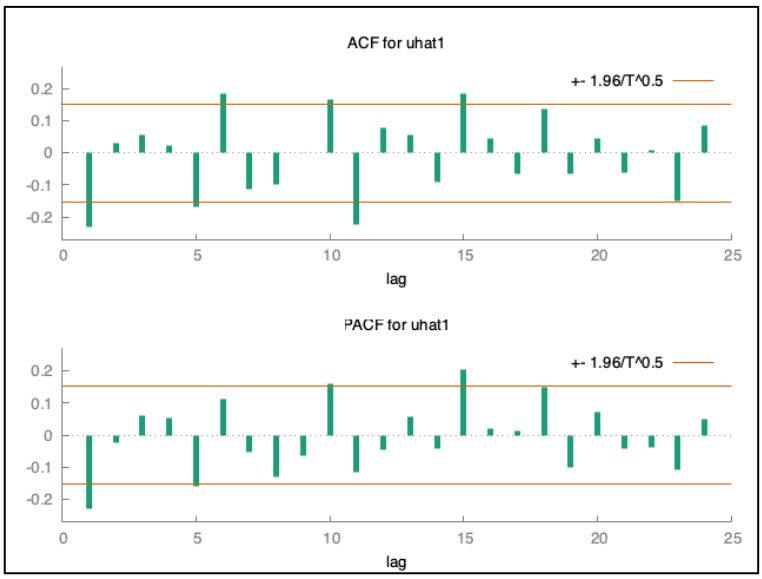
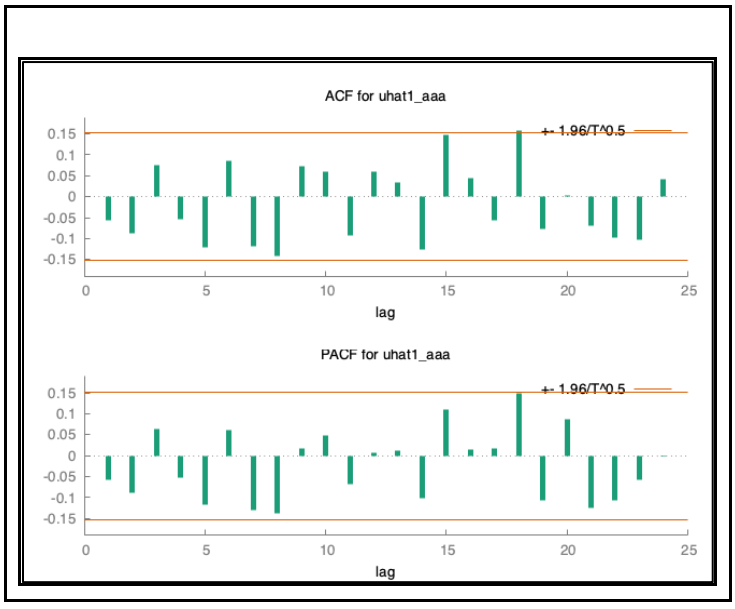


Figure A4. VAR(1) solar residual correlogram.



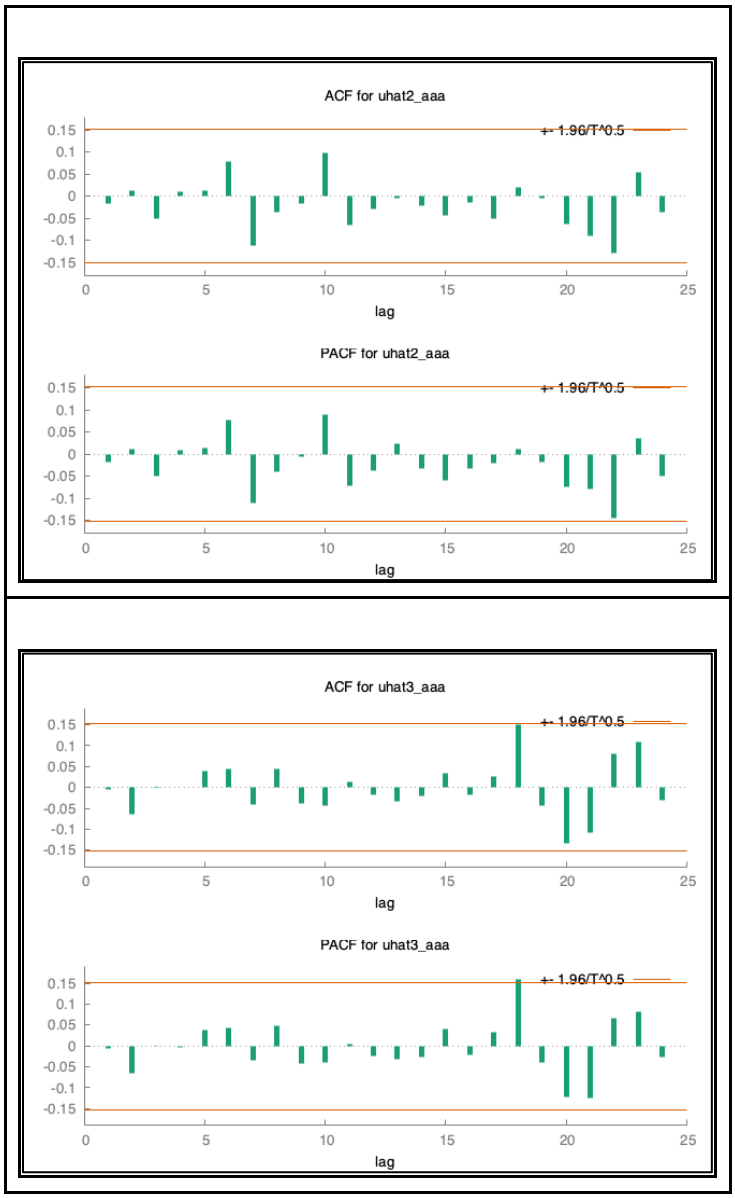


Figure A5. VAR(2) residual correlograms: solar (top), NatGas (middle), ElecPrice (bottom).

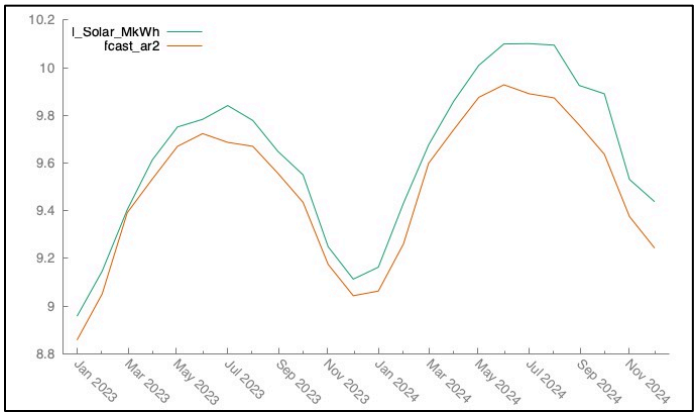


Figure A5. AR(2) in-simple forecast vs. actual log solar generation, January 2023 – December 2024.

```
VAR system, lag order 2
OLS estimates, observations 2011:03-2024:12 (T = 166)
Log-likelihood = 995.75344
Determinant of covariance matrix = 1.2368963e-09
AIC = -11.3103
BIC = -10.2417
HQC = -10.8765
Portmanteau test: LB(41) = 368.159, df = 351 [0.2539]

Equation 1: l_Solar_MkWh

                      coefficient   std. error   t-ratio   p-value
  ---------------------------------------------------------------------
  const               -0.0178604    1.36033     -0.01313  0.9895
  l_Solar_MkWh_1       0.691208     0.0737764     9.369    1.21e-16 ***
  l_Solar_MkWh_2       0.262448     0.0707762     3.708    0.0003   ***
  l_NatGas_MkWh_1      0.272221     0.161881      1.682    0.0948   *
  l_NatGas_MkWh_2     -0.244652     0.159920     -1.530    0.1282
  l_ElecPrice_cK~_1   -0.151933     0.656680     -0.2314   0.8174
  l_ElecPrice_cK~_2    0.0788371    0.653516      0.1206   0.9041
  S1                   0.140276     0.0463295     3.028    0.0029   ***
  S2                   0.369843     0.0487462     7.587    3.45e-12 ***
  S3                   0.588830     0.0536919    10.97     8.02e-21 ***
  S4                   0.418655     0.0575914     7.269    1.98e-11 ***
  S5                   0.388599     0.0467491     8.312    5.76e-14 ***
  S6                   0.264906     0.0609087     4.349    2.54e-05 ***
  S7                   0.150355     0.0760549     1.977    0.0499   **
  S8                   0.122513     0.0771670     1.588    0.1145
  S9                   0.0733738    0.0650700     1.128    0.2613
  S10                  0.0937744    0.0529480     1.771    0.0786   *
  S11                 -0.0815112    0.0416492    -1.957    0.0522   *
  time                 0.000848536  0.000881079   0.9631   0.3371

Mean dependent var   8.051552   S.D. dependent var   1.389902
Sum squared resid    1.138984   S.E. of regression   0.088024
R-squared            0.996427   Adjusted R-squared   0.995989
F(18, 147)           2277.324   P-value(F)           4.7e-170
rho                 -0.057229   Durbin-Watson        2.107526

F-tests of zero restrictions:

All lags of l_Solar_MkWh        F(2, 147) =   1327.7 [0.0000]
All lags of l_NatGas_MkWh       F(2, 147) =   1.4701 [0.2333]
All lags of l_ElecPrice_cKwh    F(2, 147) = 0.089545 [0.9144]
All vars, lag 2                 F(3, 147) =   5.1564 [0.0020]

Equation 2: l_NatGas_MkWh

                      coefficient  std. error   t-ratio  p-value
  ---------------------------------------------------------------------
  const                2.75750     0.674200      4.090   7.08e-05 ***
  l_Solar_MkWh_1      -0.0483356   0.0365646    -1.322   0.1882
  l_Solar_MkWh_2       0.0336980   0.0350777     0.9607  0.3383
  l_NatGas_MkWh_1      0.727834    0.0802307     9.072   6.97e-16 ***
  l_NatGas_MkWh_2      0.0464803   0.0792588     0.5864  0.5585
```

```
  l_ElecPrice_cK~_1   -0.905095    0.325460     -2.781   0.0061   ***
  l_ElecPrice_cK~_2    0.856221    0.323892      2.644   0.0091   ***
  S1                  -0.00855272  0.0229616    -0.3725  0.7101
  S2                  -0.136227    0.0241593    -5.639   8.50e-08 ***
  S3                  -0.0118150   0.0266105    -0.4440  0.6577
  S4                  -0.0966897   0.0285431    -3.387   0.0009   ***
  S5                   0.0527920   0.0231695     2.279   0.0241   **
  S6                   0.169413    0.0301872     5.612   9.66e-08 ***
  S7                   0.263841    0.0376939     7.000   8.48e-11 ***
  S8                   0.0781591   0.0382451     2.044   0.0428   **
  S9                  -0.134486    0.0322496    -4.170   5.18e-05 ***
  S10                 -0.152617    0.0262418    -5.816   3.62e-08 ***
  S11                 -0.174795    0.0206419    -8.468   2.35e-14 ***
  time                 0.00130639  0.000436675   2.992   0.0033   ***

Mean dependent var   11.59530   S.D. dependent var   0.252243
Sum squared resid    0.279773   S.E. of regression   0.043626
R-squared            0.973351   Adjusted R-squared   0.970088
F(18, 147)           298.2836   P-value(F)           5.3e-106
rho                 -0.016904   Durbin-Watson        2.014244

F-tests of zero restrictions:

All lags of l_Solar_MkWh        F(2, 147) =   1.6721 [0.1914]
All lags of l_NatGas_MkWh       F(2, 147) =   119.37 [0.0000]
All lags of l_ElecPrice_cKwh    F(2, 147) =   3.8698 [0.0230]
All vars, lag 2                 F(3, 147) =   2.6896 [0.0485]

Equation 3: l_ElecPrice_cKwh

                      coefficient   std. error   t-ratio   p-value
  ---------------------------------------------------------------------
  const                0.265566     0.171662      1.547    0.1240
  l_Solar_MkWh_1      -0.00324198   0.00930996   -0.3482   0.7282
  l_Solar_MkWh_2       0.000223014  0.00893137    0.02497  0.9801
  l_NatGas_MkWh_1     -0.0177034    0.0204281    -0.8666   0.3876
  l_NatGas_MkWh_2      0.00217741   0.0201806     0.1079   0.9142
  l_ElecPrice_cK~_1    0.815858     0.0828676     9.845    7.04e-18 ***
  l_ElecPrice_cK~_2    0.146345     0.0824682     1.775    0.0780   *
  S1                   0.0152743    0.00584639    2.613    0.0099   ***
  S2                   0.0220354    0.00615136    3.582    0.0005   ***
  S3                   0.00120082   0.00677547    0.1772   0.8596
  S4                  -0.00520961   0.00726756   -0.7168   0.4746
  S5                   0.0207479    0.00589935    3.517    0.0006   ***
  S6                   0.0582251    0.00768616    7.575    3.68e-12 ***
  S7                   0.0426630    0.00959749    4.445    1.72e-05 ***
  S8                   0.0218947    0.00973783    2.248    0.0260   **
  S9                   0.00582861   0.00821129    0.7098   0.4789
  S10                 -0.0298320    0.00668159   -4.465    1.59e-05 ***
  S11                 -0.0155076    0.00525578   -2.951    0.0037   ***
  time                 0.000224241  0.000111185   2.017    0.0455   **

Mean dependent var   2.380176   S.D. dependent var   0.093077
Sum squared resid    0.018138   S.E. of regression   0.011108
R-squared            0.987311   Adjusted R-squared   0.985758
```

```
F(18, 147)           635.4543   P-value(F)           1.2e-129
rho                 -0.005724   Durbin-Watson        1.991181

F-tests of zero restrictions:

All lags of l_Solar_MkWh        F(2, 147) =  0.82651 [0.4396]
All lags of l_NatGas_MkWh       F(2, 147) =  0.80741 [0.4480]
All lags of l_ElecPrice_cKwh    F(2, 147) =   820.22 [0.0000]
All vars, lag 2                 F(3, 147) =   1.0523 [0.3714]

For the system as a whole:

  Null hypothesis: the longest lag is 1
  Alternative hypothesis: the longest lag is 2
  Likelihood ratio test: Chi-square(9) = 29.8125 [0.0005]

  Comparison of information criteria:
  Lag order 2: AIC = -11.3103, BIC = -10.2417, HQC = -10.8765
  Lag order 1: AIC = -11.2391, BIC = -10.3393, HQC = -10.8739
```

Table A1. VAR(2) full estimation output.